

מבוא למדעי כדור הארץ – סיסמיקה וכבידה, תרגיל 3

1. קראו עמודים 111-113 בספר של Fowler וענו על השאלה הבאה:

During a microearthquake survey in central Turkey, an earthquake was recorded by three seismometers (Fig. 4.53). Calculate the origin time for this earthquake and locate the epicentre relative to the seismometers by using information in the table that follows.

	Hours	Minutes	Seconds
<i>Seismometer 1</i>			
P-wave arrival time	13	19	58.9
S-wave arrival time	13	20	4.7
<i>Seismometer 2</i>			
P-wave arrival time	13	20	2.6
S-wave arrival time	13	20	10.8
<i>Seismometer 3</i>			
P-wave arrival time	13	19	54.5
S-wave arrival time	13	19	57.4

Assume that the focus was at the surface and use P- and S-wave velocities of 5.6 and 3.4 km s^{-1} , respectively.

• 1 • 2

3
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50 km

Figure 4.53. A map showing the locations of seismometers 1, 2 and 3.